

Student Dietary Habits & Lifestyle Outcomes

IE 6200 · Northeastern University · ACHA 2025 Dataset · Group #2

n = 12 months

OVERVIEW

AVG PIZZA/MONTH

6.16×

Range: 4–9

AVG SLEEP QUALITY

6.78/10

Moderate – declining

AVG STRESS LEVEL

6.39/10

Rising over academic yr

ALCOHOL FREQUENCY

58.6%

Students drinking monthly

ANOVA F-VALUE

45.85

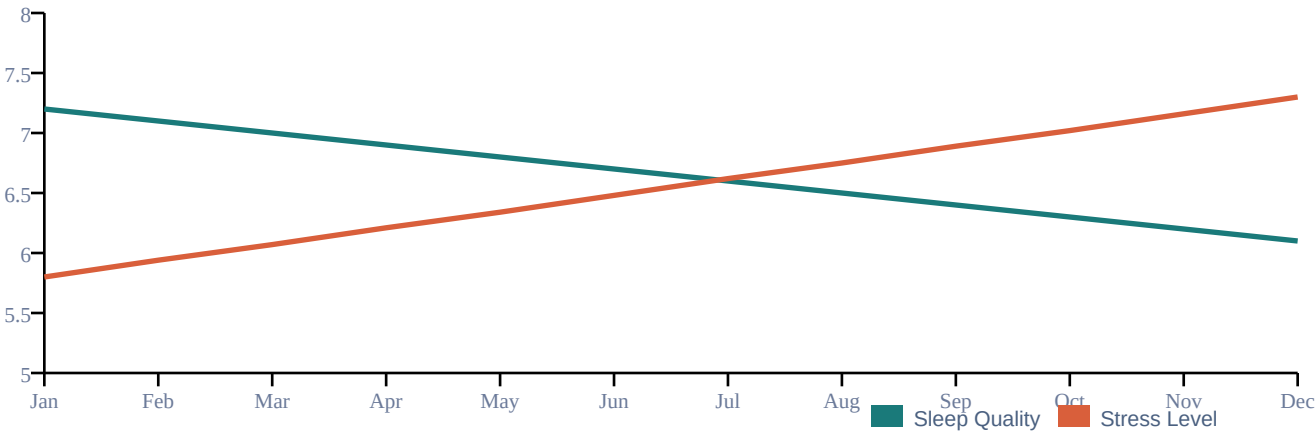
Alcohol→Stress (p=0)

DESCRIPTIVE STATISTICS

Variable	Mean	Std Dev	Min	Max
Pizza (times/mo)	6.16	1.52	4	9
Fried Chicken (times/mo)	5.16	0.91	4	7
Sleep Quality (1–10)	6.78	0.32	6.1	7.2
Stress Level (1–10)	6.39	0.46	5.8	7.3
Alcohol Frequency (%)	58.58	5.53	50	67
Binge Drinking (%)	20.92	2.91	17	26

SLEEP QUALITY VS STRESS — ACADEMIC YEAR

Monthly Sleep Quality & Stress Levels (Jan–Dec 2025)



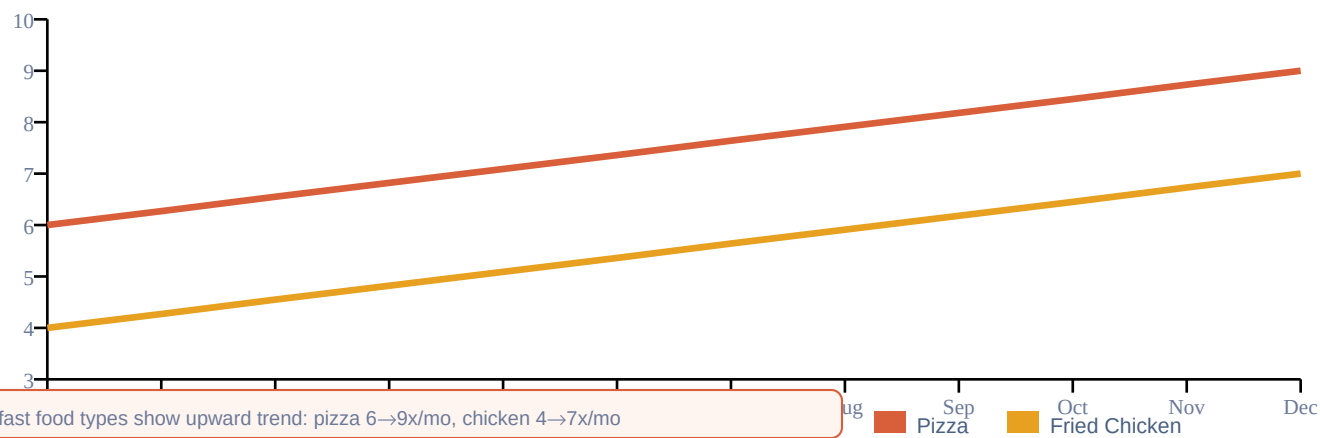
Student Dietary Habits & Lifestyle Outcomes

IE 6200 · Northeastern University · ACHA 2025 Dataset · Group #2

n = 12 months

FAST FOOD ANALYSIS

Monthly Fast Food Consumption Frequency (Jan-Dec)



REGRESSION INSIGHTS

Test	Variables	F-Value	p-Value	Result
SLR — Pizza	Pizza → Sleep Quality	2.10	0.178	Not Significant
SLR — Chicken	Fried Chicken → Sleep	2.37	0.155	Not Significant

Both simple linear regressions show weak negative trends between fast food consumption and sleep quality, but neither reached statistical significance at alpha=0.05. The relationship becomes significant only within the multiple regression framework when other dietary factors are controlled.

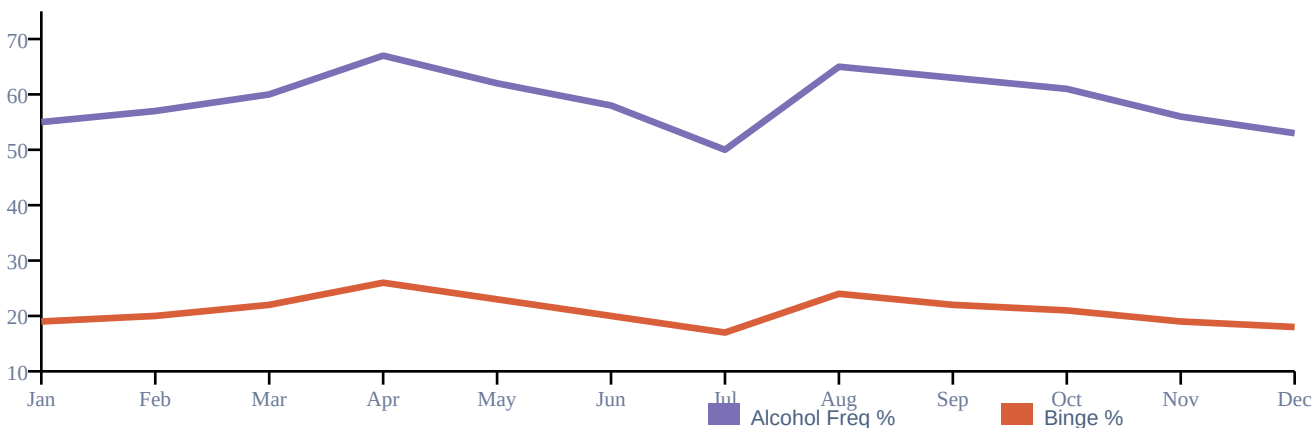
Student Dietary Habits & Lifestyle Outcomes

IE 6200 · Northeastern University · ACHA 2025 Dataset · Group #2

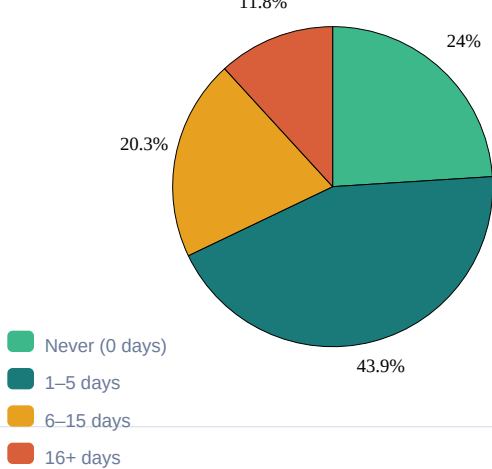
n = 12 months

ALCOHOL CONSUMPTION ANALYSIS

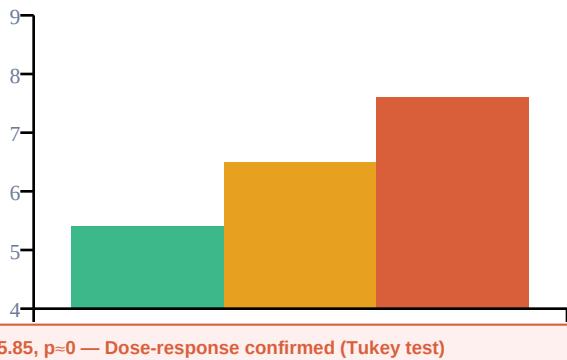
Alcohol Frequency & Binge Drinking — Monthly (%)



Drinking Frequency Distribution



Mean Stress by Alcohol Group (ANOVA)



Alcohol frequency peaks in April (67%) and August (65%), aligning with end-of-semester stress periods. The lowest consumption (50%) occurs in July during summer break. One-way ANOVA confirms that all three alcohol consumption groups (Low/Moderate/High) have significantly different mean stress levels ($F=45.85$, $p=0$), with a clear dose-response pattern.

Student Dietary Habits & Lifestyle Outcomes

IE 6200 · Northeastern University · ACHA 2025 Dataset · Group #2

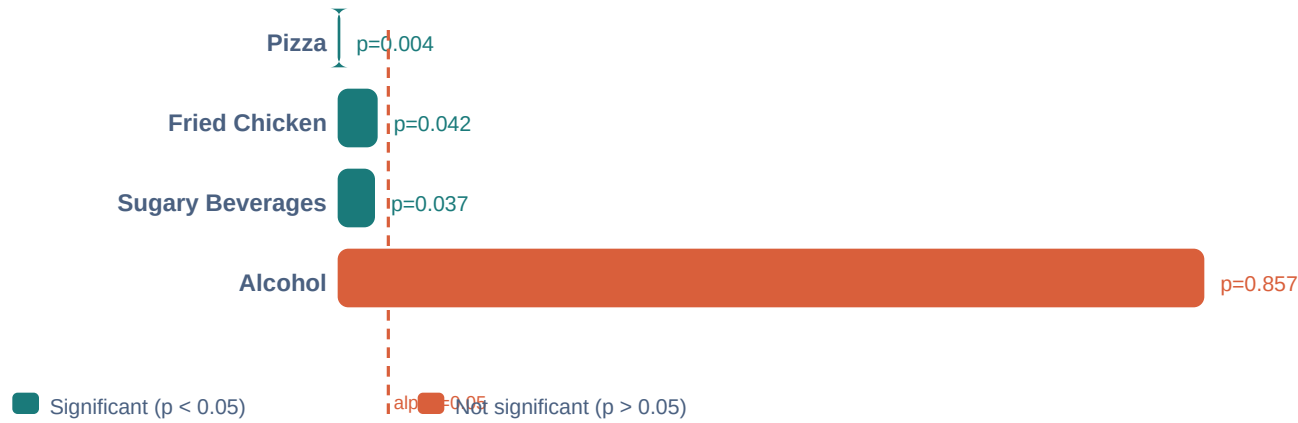
n = 12 months

STATISTICAL TESTS SUMMARY

Test	Name	Variables	F-Value	p-Value	Result
Test 1	Simple Linear Regression	Pizza/Chicken → Sleep	2.10/2.37	0.178/0.155	Not Significant
Test 2	One-Way ANOVA	Alcohol Groups → Stress	45.85	~0	Highly Significant ***
Test 3	Multiple Linear Regression	All Dietary → Stress	129.65	~0	Overall Significant ***

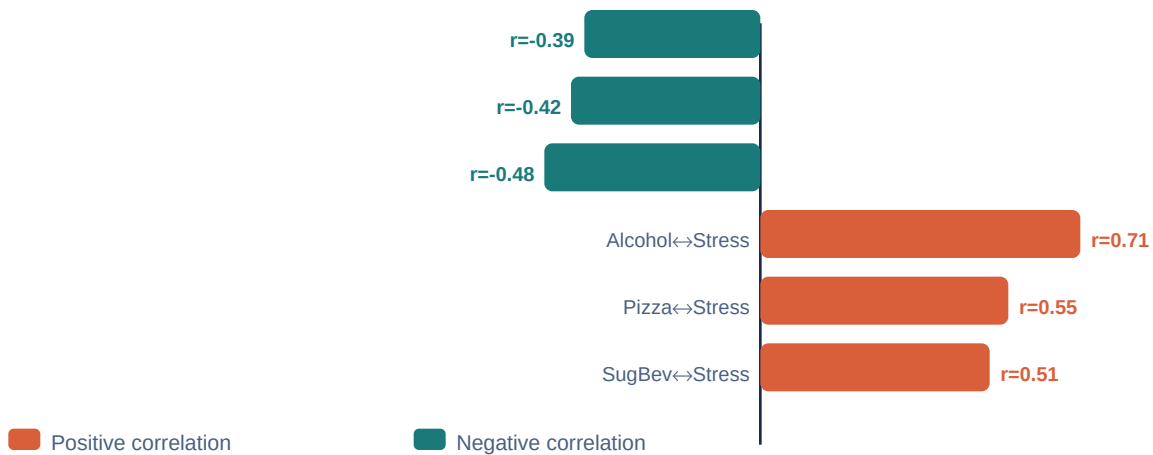
MULTIPLE REGRESSION — PREDICTOR SIGNIFICANCE

P-Values of Dietary Predictors on Stress (alpha = 0.05)



CORRELATION ANALYSIS

Correlation Strength — All Variable Pairs



Key Takeaway

The strongest relationship is Alcohol ↔ Stress ($r = 0.71$), confirmed by one-way ANOVA ($F=45.85$, $p=0$).

Fast food shows weaker but consistent negative effects on sleep quality.

In the MLR model, pizza emerges as the strongest dietary predictor of stress, while alcohol's individual effect becomes masked by correlated predictors.